12+ page PDF field guide

DISEASE & PEST MANAGEMENT

A field guide to Varroa, foulbrood, Nosema, small hive beetle, wax moth, and practical biosecurity.

Who this is for
Beekeepers, inspectors, and apiary staff responsible for colony health, diagnosis, and treatment planning.

Guide summary
Healthy colonies are protected by early detection, biosecurity, and disciplined thresholds, not by panic treatment. This field guide focuses on the main honey bee diseases and pests, how to recognise them, how to sample for them, and how to build a response plan that protects both colony health and product quality.

BeeYield brand standard: every page of this PDF carries BeeYield branding, page numbering, and a repeatable working format so the guide can be used in the yard, classroom, or packing room.

How to use this guide

Read the main sections in order, then use the worksheets and tables as operational tools. This document is written for field use, so the recommended action is always more important than memorising definitions.

Learning outcomes

- Recognise the major disease and pest risks affecting managed colonies.
- Use sampling and inspection routines before choosing treatment.
- Apply integrated pest management instead of relying on one cure.
- Strengthen biosecurity so sick hives do not become yard-wide problems.

Field Triage Guide

Varroa	Falling strength, virus pressure, visible mites in sampling	Run a standardised sample and compare to threshold
American foulbrood	Ropy brood and foul smell	Isolate equipment and escalate immediately
European foulbrood	Twisted or discoloured larvae	Confirm diagnosis and review nutrition and stress
Nosema	Poor build-up and dysentery signs	Review stress, nutrition, and colony history
Small hive beetle	Larvae in weak colonies and sliming risk	Reduce weakness and clean infested equipment
Wax moth	Webbing and damage in weak boxes	Remove weak, unused comb and improve colony strength

1. Health management starts with biosecurity and colony strength

Most pest and disease problems become expensive because they are noticed late and moved around early. Shared tools, unlabelled equipment, drifting bees, and weak quarantine habits turn one bad colony into a yard problem. Biosecurity is therefore not bureaucracy. It is the cheapest layer of disease control.

Colony strength also matters because weak colonies are easier for pests to overwhelm. Small hive beetle, wax moth, robbing pressure, and nutritional stress all hit harder when adult bee numbers are low or brood rearing has already stalled. A beekeeper who manages for strong colonies prevents many secondary problems before treatment is even discussed.

The practical rule is simple: inspect, separate, label, and clean before moving anything between hives.

Field actions

- Quarantine suspicious equipment immediately.
- Clean hive tools between suspect colonies.
- Keep yard records so problem hives can be traced.
- Support colony strength through nutrition and space management.

2. Varroa management begins with standardised sampling

Varroa is dangerous not only because mites feed on bees, but because they amplify virus pressure and weaken colonies over time. The key mistake is guessing. A beekeeper cannot manage Varroa reliably without a standard sampling routine. The Bee Health Extension guidance emphasises this clearly: know the mite level before choosing a response.

Sampling should be done in a consistent way, with enough bees and at a timing that reflects colony risk. One sample does not define the whole season, so monitoring has to be repeated through the year. Thresholds may vary by region and production goal, but the principle does not change: treatment without measurement wastes money, drives residue concerns, and teaches the beekeeper nothing.

Good Varroa management is therefore a calendar, a sampling method, and a follow-up check after action is taken.

Field actions

- Use one repeatable sampling method across the operation.
- Record mite levels by hive or by yard, not from memory.
- Re-check after treatment instead of assuming success.
- Combine genetics, monitoring, and timing rather than chasing one chemical answer.

3. Foulbrood and brood diseases demand discipline

Brood diseases punish sloppy handling. American foulbrood in particular is a serious bacterial disease that can spread through equipment, honey, and beekeeper movement. Suspicious brood patterns, foul odour, or ropy larval remains should trigger containment and expert confirmation, not casual frame swapping.

European foulbrood can look different and is often tied more closely to stress and nutrition, but it still requires careful diagnosis. The lesson is that brood problems should be inspected in good light, with frame-by-frame attention, and with clear notes. Guesswork is dangerous because the wrong response can spread a severe disease faster than the disease itself.

Operations need a written rule for when a colony is isolated, who is notified, and what can or cannot be reused.

Field actions

- Do not move brood or tools freely from a suspicious hive.
- Escalate serious brood disease signs immediately.
- Use clean diagnostic notes rather than vague impressions.
- Protect neighbouring colonies by acting early.

4. Nosema, stress, and nutrition are often linked

Not every failing colony is driven by a dramatic visible pest. Nosema and chronic stress can show up as poor build-up, weak spring performance, queen problems, or short adult bee life. These cases are frustrating because the hive may never present one dramatic clue.

The beekeeper should review the wider picture: forage continuity, feed access, moisture, crowding, travel stress, and whether the colony has been under repeated pressure from mites or disease. Stress does not excuse disease, but it often explains why one yard collapses while another with similar genetics copes better.

A health program that ignores nutrition and stress usually spends too much on treatment and too little on prevention.

Field actions

- Review stress and nutrition alongside disease signs.
- Look for patterns across yards, not just single weak hives.
- Support recovery with feed and environmental correction when needed.
- Do not assume every weak colony needs the same treatment.

5. Small hive beetle and wax moth exploit weak management

Small hive beetle and wax moth are often symptoms of weak colonies, neglected comb, or poor storage rather than isolated enemies. The Bee Health guidance on distinguishing beetle larvae from wax moth larvae is valuable because the field response can differ, but both pests become more damaging when space is unmanaged and adult bee coverage is low.

Stored comb is especially vulnerable. Boxes left with old wax, debris, and no colony strength around them become breeding space for pests. In active hives, reducing excess empty space and maintaining strong populations are often the first and best controls.

This is why pest management and ordinary husbandry cannot be separated. Strong colonies and clean comb storage reduce the chance that these pests become a crisis.

Field actions

- Identify the pest correctly before acting.
- Do not leave weak colonies oversized or neglected.
- Store comb cleanly and inspect it regularly.
- Treat pest outbreaks as a management warning, not just an insect problem.

6. Build an integrated response plan and review it after every event

The most resilient operations use integrated pest management. They monitor first, maintain strong colonies, support forage and nutrition, use biosecurity, and apply treatment only when the threshold and diagnosis justify it. This keeps chemical use more targeted and helps protect long-term colony performance.

Every disease event should end with a short review: what was seen, how quickly it was caught, what moved through the yard, what treatment or quarantine was chosen, and what the next sampling date will be. Without that review, the same problem often returns in the same form.

Health management is therefore a loop: inspect, confirm, act, verify, and improve the system.

Field actions

- Monitor before treatment whenever possible.
- Use thresholds and written criteria to guide action.
- Verify whether the response actually worked.
- Improve the management system after each event.

Glossary

Biosecurity	Practices that stop diseases and pests from entering or spreading through an operation.
Threshold	The level at which treatment or intervention becomes justified.
Varroa	A parasitic mite that weakens honey bee colonies and vectors viruses.
AFB	American foulbrood, a severe bacterial disease of honey bee brood.
Nosema	A microsporidian disease associated with stress and reduced colony performance.
IPM	Integrated pest management, a layered approach using monitoring, prevention, and targeted intervention.

Disease Inspection Sheet

Record brood appearance, odour, dead larvae signs, adult bee condition, mite sample result, and whether equipment was quarantined.

Integrated Management Plan

Write the risk, threshold, treatment or non-treatment decision, follow-up date, and how biosecurity will be maintained.

References

This guide was written from BeeYield operating practice and the following external references. Review them when adapting the guide to a new crop, region, or compliance requirement.

FAO - Good beekeeping practices: practical manual on how to identify and control the main diseases of the honeybee
https://agris.fao.org/search/en/records/64746e12d2d44cfaede240ad

Bee Health - Varroa Sampling
https://bee-health.extension.org/varroa-sampling/

Bee Health - How can I tell the difference between small hive beetle larvae and wax moth larvae?
https://bee-health.extension.org/how-can-i-tell-the-difference-between-small-hive-beetle-larvae-and-wax-moth-larvae/

FAO / Bees for Development - Beekeeping and sustainability
https://www.fao.org/family-farming/detail/en/c/1755592/